

H. Queries for Number of Palindromes

time limit per test: 5 seconds
memory limit per test: 256 megabytes

You've got a string $s = s_1s_2...s_{|s|}$ of length $|s|$, consisting of lowercase English letters. There also are q queries, each query is described by two integers l_i, r_i ($1 \leq l_i \leq r_i \leq |s|$). The answer to the query is the number of substrings of string $s[l_i...r_i]$, which are palindromes.

String $s[l...r] = s_ls_{l+1}...s_r$ ($1 \leq l \leq r \leq |s|$) is a *substring* of string $s = s_1s_2...s_{|s|}$.

String t is called a *palindrome*, if it reads the same from left to right and from right to left. Formally, if $t = t_1t_2...t_{|t|} = t_{|t|}t_{|t|-1}...t_1$.

Input

The first line contains string s ($1 \leq |s| \leq 5000$). The second line contains a single integer q ($1 \leq q \leq 10^6$) — the number of queries. Next q lines contain the queries. The i -th of these lines contains two space-separated integers l_i, r_i ($1 \leq l_i \leq r_i \leq |s|$) — the description of the i -th query.

It is guaranteed that the given string consists only of lowercase English letters.

Output

Print q integers — the answers to the queries. Print the answers in the order, in which the queries are given in the input. Separate the printed numbers by whitespaces.

Examples

input	Copy
caaaba 5 1 1 1 4 2 3 4 6 4 5	
output	Copy
1 7 3 4 2	

Note

Consider the fourth query in the first test case. String $s[4...6] = \text{«aba»}$. Its palindrome substrings are: «a», «b», «a», «aba».

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

CROC-MBTU 2012, Elimination Round (ACM-ICPC)

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
365151494	Mar/03/2026 01:45	Accepted
365151183	Mar/03/2026 01:38	Accepted
365150683	Mar/03/2026 01:27	Time limit exceeded on test 5

→ Problem tags

dp hashing strings *1800

No tag edit access

→ Contest materials

Announcement